

Powerful values of $x^4 + y^4$: A corrected computational search and rank-parity observations

Revised and extended edition

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Original edition dated 15 August 2026. This revised edition incorporates material verified against primary sources, a corrected transcription of the descent-map equations, and a new rigorous height bound derived and stress-tested for the paper’s own curve model.

Abstract

We report on a computational investigation of the smallest powerful number representable as $x^4 + y^4$ with $\gcd(x, y) = 1$, a problem posed by N. Elkies on the NMBRTHRY mailing list on 17 February 2014 and resolved, for the minimality question, by Elkies and Goel (2021/2023) — whose paper we have now located and verified, and against which we reposition the present investigation in §1 and §9. We reduce the problem to the arithmetic of the elliptic curves $E_m : v^2 = u^3 - 4m^2u$ associated with congruent numbers $2m$, derive and validate an explicit descent map recovering integer solutions from rational points on E_m (correcting a transcription error in the displayed form of this map; §3), and implement a sieve for Tunnell’s necessary criterion together with both a heuristic and a rigorous (canonical-height and lattice-enumeration based) search strategy. During verification of our own results we discovered a genuine correctness bug in our rigorous-search implementation — an unchecked assumption that PARI’s `ellrank` always returns a full generator basis, which it does not — that had caused a large majority of candidates to be silently reported as exhaustively cleared when no real search had occurred. We document the bug, its diagnosis, and its fix in full. **New in this edition:** (i) an explicit, unconditional, model-specific height bound for E_m (§5.4), derived by the same elementary gcd-control method Elkies–Goel used on their own auxiliary model, replacing our earlier empirical calibration; (ii) a fully generator-free fallback search (§7.4), adapted from Elkies–Goel’s own “Strategy 2,” for exactly the candidates our bug-fix sorted into the `partial/none/skipped` categories; and (iii) a corrected exponent in the Gaussian-integer parametrization used by that method. We also report two new explicit large solutions (at $m = 113$ and $m = 2833$), an unconditional proof — via the classical Birch–Stephens root-number formula — that every candidate surviving our sieve has analytic rank at least 2, a correction of an earlier claim that algebraic rank is uniformly 2 across this family, and a systematic $H = 10^4$ survey across $m \in [17, 10^7]$ finding 113 solutions across 14 values of m . **Further corrected in this printing,** following an independent review: the height bound of (i) is now connected rigorously to N (§5.5), lowering the honestly-provable figure from an earlier empirically-derived $N \sim 1.5 \times 10^{69}$ to 1.56×10^{58} uniformly across our candidate pool; a second, independent rigor gap is acknowledged in the generator-count check underlying Table 3 (it does not establish saturation of the Mordell–Weil basis, §5.3); the bibliographic details of [3] are corrected; and we address, with evidence, a question of whether the $m = 113$ solution duplicates prior work (it does not, §6.3).

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1 Introduction

A positive integer N is *powerful* if in its prime factorization $N = \prod_i p_i^{e_i}$ every exponent satisfies $e_i \geq 2$; equivalently, $N = a^2b^3$ for positive integers a, b with b squarefree. On 17 February 2014, N. Elkies posed the following question on the NMBRTHRY mailing list, subsequently discussed on MathOverflow:

What is the smallest powerful number that can be written as $x^4 + y^4$ with $\gcd(x, y) = 1$?

The coprimality condition is essential: since $(cx)^4 + (cy)^4 = c^4(x^4 + y^4)$, any integer can be made

powerful by multiplying by a fourth power. Elkies exhibited

$$\begin{aligned} N_1 &= 427511122^4 + 1322049209^4 \\ &= 3088257489493360278725196965477359217 = 17^3 \cdot 73993169^2 \cdot 338837713^2, \end{aligned} \tag{1}$$

and asked whether it is smallest. This was resolved affirmatively by Elkies and Goel [3], who also exhibited a second solution

$$\begin{aligned} N_2 &= 572132418369898^4 + 988478679472373^4 \\ &= 1061853595348370798528584585707993395597624934311961270177857 \end{aligned} \tag{2}$$

(again with $m = 17$ in the notation of Section 2), with computational evidence that N_2 is second-smallest.

[New in this revision] Positioning relative to Elkies–Goel. The original edition of this report cited [3] on the strength of material “supplied during this session” and flagged its bibliographic details as unverified. We have since located and read the actual paper, and it changes how the present investigation should be described. Elkies–Goel do not merely conjecture N_1 minimal: they prove it, and prove it is the *unique* solution with $N \leq 3.6125 \times 10^{37}$, for *every* squarefree $b \leq 5 \times 10^{11}$ (relying on a Tunnell/theta-coefficient computation to $2b < 10^{12}$ due to Hart, Tornara, and Watkins [6]). Their own coverage already exceeds the $m \leq 2.2 \times 10^{11}$ target quoted in Section 4.4 below — so the minimality question itself is *already closed*, unconditionally, and has been since 2021.

What the present investigation actually contributes, and should be understood as contributing, is different and complementary:

- (a) an independent re-derivation and re-verification of the reduction, descent map, and root-number theorem below, which is of some value in its own right given that our own pipeline was found to contain a serious bug (§5.3) — independent replication is exactly the kind of check that would have caught it sooner;
- (b) a much deeper (though narrower) per-curve search, to canonical height $H = 160$ (rigorously N up to 1.56×10^{58} uniformly, per the corrected bound of §5.5; substantially more for small m) and, in §10, to $H = 10^4$, far beyond anything Elkies–Goel’s minimality proof required; and
- (c) as a direct consequence of (b), two new explicit solutions ($m = 113$, $m = 2833$, §6.3) and a density survey (§10) that lie entirely outside the scope of their paper.

We adopt this framing throughout the revision below, and return to it in §9.

This report documents an independent computational investigation, built up over an extended interactive session. Beyond the search results themselves, we place particular emphasis on a serious correctness bug discovered midway through the investigation, its complete diagnosis, and its fix — because the scope of what can honestly be claimed about our own coverage changed substantially as a result, and because the diagnostic process itself produced a mathematically interesting byproduct (Section 8).

2 Reduction to Elliptic Curves

2.1 Powerful numbers and the associated curve

Any powerful N writes uniquely as $N = a^2b^3$ with b squarefree; setting $m = b$, $z = ab$ gives

$$N = mz^2, \quad m \mid z, \quad m \text{ squarefree.} \quad (3)$$

If $N = x^4 + y^4$ with $\gcd(x, y) = 1$, then since $z \geq m$ we have $N \geq m^3$, so $m \leq N^{1/3}$. A short congruence argument (using $x^4 + y^4 \equiv 1$ or $2 \pmod{16}$ depending on the parities of x, y , and that a powerful number cannot be $\equiv 2 \pmod{16}$) shows m must be odd. If an odd prime p divides $x^4 + y^4$ with $\gcd(x, y) = 1$, then -1 is a fourth power mod p , forcing $p \equiv 1 \pmod{8}$.

Proposition 2.1. *If $x^4 + y^4 = mz^2$ is powerful with $\gcd(x, y) = 1$, m squarefree, $m \mid z$, then m is odd, squarefree, and every prime factor of m is $\equiv 1 \pmod{8}$.*

2.2 The congruent-number curve

Given a solution of $x^4 + y^4 = mz^2$, set

$$u = -\frac{4x^2y^2}{z^2}, \quad v = \frac{4xy(x^4 - y^4)}{z^3}. \quad (4)$$

Subject only to $mz^2 = x^4 + y^4$, we verified by direct algebraic substitution — and independently re-verified with exact rational arithmetic on the actual N_1 solution as part of this revision — that

$$v^2 = u^3 - 4m^2u \quad (5)$$

identically. This defines the elliptic curve E_m , a twist of the curve classically associated with the congruent-number problem for $n = 2m$. Since $x, y \neq 0$ and (for $x \neq y$) $u \notin \{0, \pm 2m\}$, the resulting point is non-torsion, giving an unconditional proof that $2m$ is congruent whenever a solution exists for that m . Using the known solution (1)–(2), we independently reconfirmed that $n = 34$ ($m = 17$) is congruent this way – via an explicit non-torsion point, not merely via Tunnell’s criterion.

2.3 Tunnell’s criterion as a necessary filter

By the Coates–Wiles theorem (the CM special case of Gross–Zagier–Kolyvagin), if E_{2m} has positive rank then $L(E_{2m}, 1) = 0$. Tunnell’s theorem [2] gives an efficiently computable criterion, in terms of counts of representations of $n = 2m$ by two ternary quadratic forms, for this vanishing; for $n = 2m$, m odd, writing

$$A_m = \#\{(x, y, z) \in \mathbb{Z}^3 : m = 4x^2 + y^2 + 32z^2\}, \quad B_m = \#\{(x, y, z) \in \mathbb{Z}^3 : m = 4x^2 + y^2 + 8z^2\}, \quad (6)$$

$n = 2m$ congruent implies $B_m = 2A_m$. We implemented these counts as an FFT-based sieve (Section 4), validating the implementation against brute-force enumeration for small m and against the classical list of small congruent numbers, including $m = 17$ itself, whose congruence had already been established unconditionally in Section 2.2.

3 The Descent Map

The map (4) sends solutions of $x^4 + y^4 = mz^2$ to points on E_m ; the search strategy of Section 5.2 requires the inverse. From (4) and $mz^2 = x^4 + y^4$,

A note on this equation. The displayed form of the following relation in the original edition (“ $2m - u^2 = \left(\frac{x^2+y^2}{z}\right)^2$ ”, with no division by 2 and a spurious square on u) does not hold numerically — substituting the actual N_1 point gives $2m - u^2 \approx -15.6$, which cannot equal a square. This was a transcription artifact from the original document, not an error in the underlying computation: the paper’s own prose immediately below has always correctly described testing $(2m - u)/2$, $(2m + u)/2$, and $-u$ for being perfect squares. We give the corrected display form here, re-verified against the exact N_1 point (independently, using exact rational arithmetic):

$$\frac{2m - u}{2} = \left(\frac{x^2 + y^2}{z}\right)^2 =: P^2, \quad \frac{2m + u}{2} = \left(\frac{x^2 - y^2}{z}\right)^2 =: Q^2, \quad -u = \left(\frac{2xy}{z}\right)^2 =: W^2, \quad (7)$$

with $P^2 - Q^2 = W^2$ automatic, and one checks directly that $v = 2WPQ$. Given a rational point (u, v) , we test whether $(2m - u)/2$, $(2m + u)/2$, $-u$ are each perfect squares of rationals; if so we recover signed square roots P, Q, W consistent with $v = 2WPQ$ (a generic 4-fold sign ambiguity, corresponding to $(x, y) \mapsto (\pm x, \pm y)$, $x \leftrightarrow y$), set $A = (P + Q)/2 = x^2/z$, $B = (P - Q)/2 = y^2/z$, recover $x : y$ as $\sqrt{A/B}$, and clear denominators via the scaling freedom $(x, y, z) \mapsto (cx, cy, c^2z)$ to obtain a minimal integer solution. Most points of $E_m(\mathbb{Q})$ fail the perfect-square conditions and correctly yield no solution — expected, since (4) realizes one specific torsor for E_m among several inequivalent ones.

Proposition 3.1 (Validation). *Applied to the point on E_{34} obtained from (1) via (4), this map recovers $(x, y, z) = (427511122, 1322049209, 426218494746902449)$ exactly (up to sign/swap). Applied to a point independently located via lattice-point search (Section 5.2), it recovers (2) exactly.*

This was checked with exact rational arithmetic throughout, not floating point — and, as part of this revision, independently re-checked from scratch in a fresh Python implementation using exact fractions, with identical results.

4 The Sieve: an FFT-Based Evaluation of Tunnell’s Criterion

4.1 Method

The counts (6), for all m up to a bound X simultaneously, are coefficients of a triple convolution of theta series. We compute the convolution of the two cheaper terms directly via a sparse outer-product/bincount ($O(X)$ work), then convolve the resulting dense array with the sparse third-term series via FFT, for $O(X \log X)$ overall time. A separate arithmetic sieve (odd m , squarefree, all prime factors $\equiv 1 \pmod{8}$) is combined with the Tunnell counts to produce the final candidate list.

4.2 Validation

Checked against brute-force counting for numerous individual m (exact agreement including non-trivial counts such as $(A, B) = (1252, 2556)$ and $(336, 672)$ at six-digit test points), and against the classical table of small congruent numbers together with the independently-confirmed $m = 17$ case above.

4.3 Memory, and a practical improvement

An initial implementation (32-bit counts, single-precision FFT, power-of-two FFT lengths) required substantially more memory than a naive $O(X)$ estimate suggests, owing to FFT padding: at $X = 3 \times 10^7$, peak memory was 2.7 GB, with a hard ceiling near $X \approx 3.4 \times 10^7$ at the power-of-two padding boundary. Replacing power-of-two lengths with the smallest 5-smooth length at least $2X$ (`scipy.fft.next_fast_len`) removed most of this overhead: $X = 6 \times 10^7$ needed only 2.6 GB, and the practical ceiling extended to $X = 8.8 \times 10^7$ at 3.7 GB – roughly 42 bytes of peak memory per unit of X , versus ≈ 90 bytes/unit under the power-of-two implementation.

4.4 Extrapolation to the full range, and an existing alternative

Combined with Elkies’s observation that $x^4 + y^4 = m^3$ has no coprime solutions (so the smallest admissible m is 17, matching N_1 ’s own factorization), the target range for an unconditional resolution of a search built entirely from scratch is m up to roughly 2.2×10^{11} .¹ Extrapolating the smooth-FFT measurements under $O(X \log X)$ scaling gives an estimated 9–10 TB of memory and roughly 3 days of single-threaded compute for the full range – a specialized-hardware requirement, not available in the environment used for this investigation. We also benchmarked PARI’s `qfrep` as a possible alternative; it scaled markedly worse than our sieve at matched sizes and was not pursued further. We separately tested PARI’s `ellrootno` as a possible cheap rank-parity pre-filter; it returned +1 uniformly regardless of whether a candidate was Tunnell-positive, so it does not discriminate and is not useful as a filter (though its uniformity is itself explained in Section 8).

[New in this revision] As noted in §1, this 9–10 TB estimate is a statement about extending *our own* implementation, not about the state of the art: Hart, Tornara and Watkins [6] computed the analogous theta-coefficient sieve to $2b < 10^{12}$ – beyond both our current reach and our stated target – using a disk-based, multimodular (i.e. NTT-style, rounding-error-free) out-of-core polynomial multiplication designed specifically to avoid this memory wall. Before investing further engineering in a segmented or distributed rewrite of our own sieve (as proposed in §9.1(a) of the original edition), it would be more efficient to determine whether the Hart–Tornara–Watkins data or code is obtainable, or at minimum to benchmark a rewrite against their reported memory/time figures rather than extrapolating blind from our own 8.8×10^7 data point.

¹**[New in this revision]** We have not located an explicit derivation of this specific figure in our own working notes, and flag that it should be shown or replaced. For comparison, the closely related figure that Elkies–Goel *do* derive explicitly (§1) is $b \leq 5 \times 10^{11}$, matching their available Tunnell data via [6] at $2b < 10^{12}$; our own 2.2×10^{11} may be an independent, more conservative estimate from an earlier stage of this project, but we cannot presently reconstruct which bound on N it was meant to correspond to, and list this as an open item in §9.1.

5 Search Methodology

For each sieve-surviving m , we computed rank and generators of $E_m(\mathbb{Q})$ via `ellrank` in PARI/GP 2.15.4.

5.1 Heuristic search

Initially we searched each curve by enumerating integer combinations $aG_1 + bG_2$ of the generators with $|a|, |b| \leq d$ for fixed depth d (typically 4 or 8), descending each resulting point and checking coprimality and $m \mid z$. This has no rigorous stopping criterion – a negative result only shows no solution among the finitely many points tested. Its results are reported in Section 6 (it is how the $m = 113$ and $m = 2833$ solutions were found).

5.2 Rigorous search via canonical heights

The canonical height $\hat{h}$ on $E_m(\mathbb{Q})$ is a positive-definite quadratic form on the Mordell–Weil lattice, computable via a Gram matrix of pairwise heights (`ellheight`); all lattice points up to a height bound H can then be enumerated exhaustively via the Fincke–Pohst algorithm (`qfminim`). We calibrated $\hat{h}(P) \approx \ln(x^4 + y^4)$ on our curve model directly, confirmed at three independent points (discrepancies of 0.09–0.23 against N_1 , N_2 , and an independently-found $m = 113$ solution), and used $H = 160$ throughout (covering N up to $\sim 1.5 \times 10^{69}$ on this empirical calibration alone).

[New in this revision] *This empirical N -figure is corrected in §5.5 below.* Connecting $H = 160$ rigorously to a bound on N – rather than via the empirical calibration above – turns out to require more care than either edition of this paper gave it; §5.5 supplies the missing step and revises the figure accordingly.

Remark 5.1 (Torsion cosets). **[New in this revision]** E_m has rational 2-torsion: the point at infinity together with $(u, v) = (0, 0)$, $(2m, 0)$, and $(-2m, 0)$ (§2.2). A lattice search via `qfminim` on the free generators G_1, G_2 enumerates integer combinations $aG_1 + bG_2$; a solution could in principle correspond to a point in a *non-identity* torsion coset, $T + aG_1 + bG_2$ for one of the three nontrivial 2-torsion points T , whose height is not simply that of $aG_1 + bG_2$ alone. We have not found an explicit statement in either edition of this paper’s own working notes of whether all four cosets were searched or only the identity coset – this needs to be confirmed (and stated explicitly) against the actual implementation before the `full` category’s exhaustiveness claim can be considered complete, independent of the saturation question above. We flag rather than resolve this here, and add it to §9.1.

5.3 A critical correctness bug, and its fix

This subsection documents an error in our own earlier work and its correction; we report it in full rather than silently revising prior claims.

Our first implementation of the rigorous search proceeded, after calling `ellrank`, to build a `qfminim` search script assuming a full rank-dimensional generator basis was available, without checking that PARI had actually returned that many generator points. `ellrank` at its default search effort frequently returns fewer generators than the (correctly, separately proven) rank: when this happened,

the generated PARI script referenced an undefined variable, silently produced no output, and our code nonetheless reported the candidate as exhaustively cleared with no solution – a false claim of rigor whenever this occurred.

We discovered this during a verification pass prompted by directly checking whether the number of generators `ellrank` returned matched the proven rank across the candidate pool, rather than assuming it did. Diagnosis proceeded in two stages:

- (i) **Rank-4 curves.** All 3,611 of the candidates our original code had marked non-exhaustive (out of 88,445 total) were confirmed to have rank exactly 4, not 2 – correctly skipped by the code (which explicitly declines ranks above 2, since `qfminim` scales poorly there), but revealing that an earlier claim of ours (rank uniformly 2 across this family) was an overclaim; see Section 8.
- (ii) **Rank-2 generator shortfall.** Re-querying `ellrank` fresh on 500 candidates randomly sampled from those our original code had reported as exhaustively cleared with no solution, all 500 confirmed rank 2, but only 13 (2.6%) had a full 2-generator basis at default effort; 127 (25.4%) had exactly 1; 360 (72.0%) had 0. Since our code proceeded regardless, roughly 97% of this (admittedly biased – see below) sample had never actually been searched.

The fix. `rigorous_search.m` now checks the number of generators returned against the proven rank before proceeding, and reports one of four outcomes: **full** (complete basis; genuinely exhaustive up to H over the whole lattice), **partial** (fewer generators than the rank; exhaustive only over the sublattice spanned by the generators found – a strictly weaker guarantee), **none** (zero generators; no search possible), or **skipped** (rank > 2 , not attempted). Only **full** counts as exhaustive. We validated the fix against all four outcome types on known cases (Table 1) before redeploying it.

Case	Old behavior	New behavior
$m = 17$	<code>exhaustive=True</code> (correct, by luck)	<code>status=full</code> , finds N_1, N_2 ✓
$m = 45187697$ (0 gens)	<code>exhaustive=True</code> (wrong)	<code>status=none</code> ✓
$m = 6800561$ (1 of 2 gens)	<code>exhaustive=True</code> (wrong)	<code>status=partial</code> ✓
$m = 23137$ (rank 4)	<code>exhaustive=False</code> (correct)	<code>status=skipped</code> ✓

Table 1: Validation of the fixed generator-count check against all four outcome types before large-scale redeployment.

We also tested whether raising `ellrank`’s search effort rescues generator-shortfall cases at scale: on 10 confirmed-shortfall candidates, raising effort from 0 to 3 improved none of them to a full basis – a materially worse rescue rate than the 1-in-5 success observed on Elkies–Goel’s own five small exceptional cases (Section 7), suggesting generator-finding difficulty grows with m , consistent with the regulator-based explanation of Section 7.2.

Remark 5.2 (A second, subtler gap in the same spirit as §5.3 – saturation). [\[New in this revision\]](#) Everything above checks that `ellrank` returns as *many* generator points as the proven rank. It does not check that those points *saturate* $E_m(\mathbb{Q})$ – that is, that they generate the full Mordell–Weil group rather than a proper finite-index subgroup of it. This distinction is not academic: PARI ships a separate function, `ellsaturation`, precisely because a rank-many set of independent points returned by a descent search need not be saturated, and its authors added it to `ellrank` shortly after that function’s own introduction for exactly this reason. If the generators used for a

full candidate span an index- k sublattice of the true Mordell–Weil lattice for some $k > 1$, then `qfminim` on that sublattice can miss genuine solutions of height $\leq H$ that happen to lie outside it – the search would still terminate and report no solution, but the “genuinely exhaustive” label in Table 3 would not be earned. We did not run `ellsaturation` on the 975 full candidates before this revision, and so we downgrade our own confidence accordingly: the full category as currently computed is exhaustive over the sublattice spanned by the reported generators, which is a necessary but not sufficient condition for exhaustiveness over $E_m(\mathbb{Q})$ itself. Closing this gap requires running `ellsaturation` (at least at small primes; a full saturation certificate additionally bounds which primes could possibly divide the index) on each full candidate and recording the resulting index bound, which we have not yet done at scale. We list this as the first item of §9.1 below.

5.4 A rigorous height bound

[New in this revision]

The original edition used the empirical calibration of §5.2 as its only height bound, and listed a rigorously derived, model-specific bound as future work (§9.1(d) below). We supply one here, obtained by the same method Elkies–Goel use for their own auxiliary model [3, Prop. 1] – an elementary gcd-control lemma combined with Tate’s telescoping argument – applied directly to E_m itself, so that no change of model is needed.

The duplication map on $y^2 = x^3 + Ax + B$ sends x -coordinate x to $\varphi(x)/\psi(x)$ with $\varphi(x) = x^4 - 2Ax^2 - 8Bx + A^2$, $\psi(x) = 4(x^3 + Ax + B)$. For E_m ($A = -4m^2$, $B = 0$):

$$\varphi(x) = (x^2 + 4m^2)^2 \quad (\text{monic, degree 4}), \quad \psi(x) = 4x^3 - 16m^2x \quad (\text{degree 3}). \quad (8)$$

Since φ, ψ are coprime, they admit a Bézout identity $F\varphi + G\psi = 1$ over $\mathbb{Q}(m)[x]$; computing it explicitly gives

$$F(x) = \frac{16m^2 - 3x^2}{256m^6}, \quad G(x) = \frac{x(20m^2 + 3x^2)}{1024m^6}, \quad (9)$$

so that $R := 1024m^6$ clears both denominators. Because φ is monic, the standard gcd-control lemma ([3, Lemma 1]; a routine resultant/Bézout argument, not requiring Silverman’s full local-heights-at-every-place machinery) gives, for $u = p/q$ in lowest terms and Φ, Ψ the homogenized numerator/denominator of the doubled coordinate,

$$\gcd(\Phi(p, q), \Psi(p, q)) \mid 1024m^6. \quad (10)$$

Combined with elementary two-sided estimates $|\Phi|, |\Psi|$ in terms of $H(u) = \max(|p|, |q|)$ (using $p^2 + 4m^2q^2 \geq H(u)^2$ for $m \geq 1$, and the triangle inequality in the other direction), one obtains, for every $m \geq 1$ and every point Q on E_m ,

$$h(x([2]Q)) - 4h(x(Q)) \in [-6 \ln m - 10 \ln 2, 2 \ln(1 + 4m^2)]. \quad (11)$$

Telescoping via Tate’s argument ($\sum_{k \geq 1} 4^{-k} = 1/3$) gives the usable form: for any non-torsion P with $x(P) = p/q$ in lowest terms,

$$-2 \ln m - \frac{10}{3} \ln 2 \leq \hat{h}(P) - \ln(\max(|p|, |q|)) \leq \frac{2}{3} \ln(1 + 4m^2). \quad (12)$$

Remark 5.3. This bound is stated in terms of $\ln(\max(|p|, |q|))$, the naive height of $u = x(P)$ itself – not $\frac{1}{2}$ of it. We checked this normalization independently: computing $\hat{h}(P)$ directly via

Tate’s doubling limit on the exact N_1 point (with no reference to PARI) gives successive estimates 84.04, 84.09, 84.10, $\dots$, converging exactly to the paper’s own reported value 84.11 for that point (Appendix A.1) – confirming that $\hat{h}(P) \approx \ln(x^4 + y^4)$ (coefficient 1) is the correct calibration *on this specific model*, and explaining why it differs from the coefficient of $\frac{1}{2}$ appearing in [3]: their bound is stated on a different, 2-isogenous auxiliary model, and canonical heights are not isogeny invariant.

Remark 5.4 (Scope of verification). The Bézout identity and the gcd-divisibility claim above were checked symbolically (exact computer algebra). The two-sided per-doubling inequality was stress-tested numerically across 20,000 random (m, p, q) triples spanning m from 17 to 100,003, with zero violations and the upper bound essentially tight (worst-case margin $\sim 10^{-13}$) – evidence against a hidden error, though this is a numerical check, not a machine-verified proof. We have not attempted to show $R = 1024m^6$ is optimal; a smaller valid R would tighten the bound further, exactly as Elkies–Goel’s own $R = 16$ need not be optimal for their model. At the target range $m \leq 2.2 \times 10^{11}$ ($\ln m \leq 26.1$), the two correction terms are at most ≈ 54.6 and ≈ 36 respectively – small relative to $H = 160$, so the bound remains usable across the full target range, not only for small m . Full derivation details, including the intermediate polynomial identities, are given in Appendix C.

5.5 From height to N : correcting the empirical figure

[New in this revision]

Section 5.2 states that $H = 160$ “covers N up to $\sim 1.5 \times 10^{69}$ ” on the strength of the empirical calibration $\hat{h}(P) \approx \ln(x^4 + y^4)$ alone. Neither edition of this paper had, until now, actually connected the height bound of §5.4 to a proved statement about N . Doing so exposes a real gap, not merely a missing formality: the correct bound turns out to license a substantially smaller N at $H = 160$ than was claimed.

Lemma 5.5. *For a primitive solution ($\gcd(x, y) = 1$) of $x^4 + y^4 = mz^2$, the fraction $u = -4x^2y^2/z^2$ is already in lowest terms: $\gcd(4x^2y^2, z^2) = 1$.*

Proof. Since m is odd (Proposition 2.1) and $N = mz^2$ is odd (§2.1), z is odd, so $2 \nmid z^2$; since exactly one of x, y is even, $4 \mid 4x^2y^2$ contributes only the prime 2 beyond x^2y^2 ’s odd part, and $2 \nmid z^2$ kills any contribution there. For an odd prime $p \mid z$: $p \mid mz^2 = x^4 + y^4$, and if p also divided x (resp. y) it would force $p \mid y^4$ (resp. x^4) via $x^4 + y^4 \equiv 0$, hence $p \mid y$ (resp. x), contradicting $\gcd(x, y) = 1$. So no odd prime divides both z and xy , and combined with the previous sentence, $\gcd(4x^2y^2, z^2) = 1$. $\square$

Consequently $H(u) := \max(4x^2y^2, z^2)$ exactly, with no further reduction. Since $4x^2y^2 \leq 2(x^4 + y^4) = 2N$ (AM–GM) and $z^2 = N/m \leq N$,

$$H(u) \leq 2N, \quad \text{so} \quad h(u) \leq \ln(2N).$$

Combining with the upper bound of equation (12) (§5.4),

$$\hat{h}(P) \leq \ln(2N) + \frac{2}{3} \ln(1 + 4m^2). \quad (13)$$

This is the missing bridge from a height cutoff to an N -bound. Solving for the largest N guaranteed to be found by a height- H search on E_m ,

$$N_{\max}(m, H) = \frac{1}{2} \exp\left(H - \frac{2}{3} \ln(1 + 4m^2)\right). \quad (14)$$

Remark 5.6 (Corrected figure). [\[New in this revision\]](#) At $H = 160$, equation (14) is *not* uniform in m the way the empirical calibration suggested: it ranges from $N_{\max}(17, 160) \approx 1.39 \times 10^{67}$ down to $N_{\max}(8.8 \times 10^7, 160) \approx 1.56 \times 10^{58}$ across our candidate pool. Since every full-classified candidate in this pool has $m \leq 8.8 \times 10^7$, the uniform, honest replacement for “covers N up to $\sim 1.5 \times 10^{69}$ ” is:

$H = 160$ rigorously covers $N \leq 1.56 \times 10^{58}$ for every candidate in our pool

(with a better, m -specific bound available for any individual smaller m , e.g. $\approx 1.4 \times 10^{67}$ at $m = 17$ itself). To rigorously recover the originally-stated $N \sim 1.5 \times 10^{69}$ uniformly across $m \leq 8.8 \times 10^7$ would require $H \geq 185.3$, not $H = 160$; equivalently, our completed $H = 160$ `qfminim` runs remain fully valid computations, but the honest scope of what they prove about N is the smaller figure above, not the originally-claimed one. We correct Proposition 6.1 and the abstract accordingly. This does not affect $m = 17$ specifically: N_1, N_2 , and the further $m = 17$ solutions of Appendix A/B are all far below even the smaller, corrected bound.

6 Results

6.1 Coverage

Table 2 summarizes candidate ranges and methods. 88,445 candidate values of m were identified by the sieve, up to $m \leq 8.8 \times 10^7$: 28,010 up to 2×10^7 , 13,361 more up to 3.3×10^7 , and 47,074 more up to 8.8×10^7 .

Pass	Coverage	Result
Heuristic (depth 4 & 8)	88,445/88,445 (100%)	$m = 17$ (N_1, N_2), $m = 113$, $m = 2833$
Rigorous, original buggy code	claimed 88,445/88,445	claim invalid – see §5.3
Rigorous, fixed code	11,461/88,445 (13.0%), in progress	see Table 3

Table 2: Coverage summary, including the withdrawn claim from the original buggy implementation.

Status	Count (of 11,461 checked)	Percentage
none (no generators; no search possible)	5,695	49.7%
partial (sublattice search only)	4,442	38.8%
full (genuinely exhaustive)	975	8.5%
skipped (rank 4)	349	3.0%

Table 3: Outcome breakdown under the fixed code, current progress. Only the **full** category (8.5% so far) carries a genuine exhaustiveness guarantee; **partial** is weaker (sublattice only), and **none** carries no search guarantee beyond the rank itself being known. [\[New in this revision\]](#) Section 7.4 below gives a method applicable to all three of the non-**full** categories.

6.2 What is, and is not, established

Proposition 6.1 (Corrected, honestly-scoped result). *Across all 88,445 sieve-surviving candidates $m \leq 8.8 \times 10^7$, a heuristic (non-exhaustive) search found solutions only at $m = 17$ (giving N_1, N_2), $m = 113$, and $m = 2833$ – all $\geq N_1$. Of these candidates, 11,461 (13.0%) have so far been re-examined under the corrected rigorous procedure; of those, 975 (8.5% of the re-examined subset) are exhaustive over the sublattice spanned by their reported generators up to $N \leq 1.56 \times 10^{58}$ (§5.5) – and, pending the saturation check of §5.3, over the full Mordell–Weil group up to that N – and among these only $m = 17$ has a solution. The remaining candidates – both the 86,984 not yet re-examined and the partial/none/skipped portion of those that have been – carry no exhaustiveness guarantee from this procedure.*

[New in this revision] Both qualifications above are corrections made in this revision: the N -bound replaces the original edition’s $N \sim 1.5 \times 10^{69}$ figure (§5.5), and the sublattice/saturation caveat reflects §5.3’s newly-added remark that our generator-count check does not by itself establish saturation. Neither correction changes the paper’s headline claim, since the actual small- m verified solutions (N_1, N_2 , and everything in Appendix A/B) sit far below 1.56×10^{58} in any case; what changes is how much can honestly be claimed about the *absence* of solutions among the candidates where none was found.

We emphasize that this proposition, not a stronger exhaustive claim, is what our computation currently supports (and, per §1, that the minimality question itself is separately and already settled unconditionally by [3] regardless of the state of this proposition). An earlier draft of this report claimed full rigorous coverage of all 88,445 candidates; that claim is withdrawn for the reasons given in Section 5.3.

6.3 New explicit solutions

Two new solutions were found via the heuristic search, both far larger than N_1 : $m = 113$ ($x^4 + y^4$ has 162 decimal digits) and $m = 2833$ (594 digits). Full values are in Appendix A. Neither is captured by the rigorous pass at $H = 160$, since both exceed that height bound. As noted in §1, these lie entirely outside the scope of Elkies–Goel’s minimality proof and stand as genuine new data regardless of it.

Remark 6.2 (On the novelty of the $m = 113$ solution). **[New in this revision]** A question raised about this revision was whether the $m = 113$ solution above might already appear in [3], rather than being new. We checked this directly against their paper rather than assuming either way. Their Theorem 1 is explicit: for $b \leq 10^4$ other than the five exceptional values $\{4721, 4777, 6497, 6577, 9881\}$ (and 113 is not among them), the *only* solutions with $x^4 + y^4 \leq 2^{2/3}e^{140} \approx 1.0044 \times 10^{61}$ occur at $b = 17$. Our $m = 113$ solution has 162 digits ($N \approx 7.5 \times 10^{161}$), and their paper’s overall computational claim (abstract) is bounded at $N \leq 3.6125 \times 10^{37}$ across all $b \leq 5 \times 10^{11}$ – their own stated “future work” is exactly *how one might go further than that bound*, which confirms they had not, at the time of writing, done so. Both of their explicit bounds are far below our $m = 113$ solution’s height, so there is no version of their published result that already contains it. We are confident the $m = 113$ solution is correctly described as new.

7 Improving Elkies–Goel’s Strategy 1

Prompted by a request to identify concrete improvements to the Elkies–Goel method, we examined the five values of m their manuscript reports as requiring a fallback strategy: $m \in \{4721, 4777, 6497, 6577, 9881\}$, each squarefree with prime factors $\equiv 1 \pmod{8}$.

[New in this revision] *A caveat added in this revision.* Having now read [3] directly, we note that their own paper describes a second, entirely generator-free method (their “Strategy 2,” §7.4 below) for exactly this kind of hard case. We do not have enough detail from our reading to say with certainty which of their two strategies was used for each of the five specific values above; the effort-tuning and BSD-triage findings of this section should therefore be read as an independently discovered, complementary approach to the generator-finding problem – one that happens to work by pushing harder on the same generator-based machinery, rather than as a reconstruction of Elkies–Goel’s own resolution of these particular cases.

7.1 Generator-finding, not rank, is the obstruction

`ellrank` gave matching rank bounds $[2, 2]$ for all five – rank is never in question. Table 4 shows the generator-finding outcome at default and raised effort.

m	Rank (proven)	Gens at default effort	Gens at raised effort
4721	$[2, 2]$	1 of 2	1 of 2 (up to effort 5)
4777	$[2, 2]$	2 of 2	—
6497	$[2, 2]$	1 of 2	2 of 2 (effort 1)
6577	$[2, 2]$	1 of 2	1 of 2 (up to effort 5)
9881	$[2, 2]$	2 of 2	—

Table 4: Generator-finding outcomes for Elkies–Goel’s five exceptional cases.

Raising effort resolved $m = 6497$ immediately at negligible cost, reducing the exceptional set from five to four with no new method. Two cases (4721, 6577) did not yield a second generator even at effort 5, nor via independent brute-force search (Python and native-PARI implementations, plus PARI’s `ellratpoints` up to naive height 10^9) – all consistent with the finding in Section 5.3 that effort-bumping is considerably less reliable at larger scale than this five-case sample alone would suggest.

7.2 A validated triage rule

For the two unresolved cases we estimated the missing generator’s canonical height via the rank-2 Birch–Swinnerton-Dyer formula

$$\frac{L''(E, 1)}{2} = \Omega_E \cdot \text{Reg}(E) \cdot \prod_p c_p \cdot \#\text{Sha}(E) / (\#E(\mathbb{Q})_{\text{tors}})^2, \quad (15)$$

using PARI’s `ellbsd` and `lfun(E, 1, 2)`, solving for the regulator under $\#\text{Sha}(E) = 1$, and approximating $\text{Reg}(E) \approx \hat{h}(G_1)\hat{h}(G_2)$ (dropping the height-pairing cross term). Both estimated heights (≈ 88.5 and ≈ 146.3) were far beyond anything reachable by direct search. Testing this estimate

on all six curves (five exceptional cases plus $m = 17$ as a control) shows a clean separation: every case where a second generator was actually found has estimated height ≤ 55 ; both cases where it was not have estimated height ≥ 88 (Table 5).

m	$\hat{h}(G_1)$	Est. $\hat{h}(G_2)$	Outcome
17 (control)	2.51	2.8	found trivially
4777	5.33	36.3	found at default effort
6497	4.44	36.0	found at effort 1
9881	5.78	54.7	found at default effort
6577	8.83	88.5	not found (effort ≤ 5)
4721	8.71	146.3	not found (effort ≤ 5)

Table 5: The BSD-based estimate, computed before further search, correctly separates tractable from intractable cases across all six curves tested.

This costs one inexpensive `ellbsd/lfun` evaluation and would let a pipeline route hard cases directly to a fallback method before investing in a doomed search. The sample (six curves) is small; a specific numerical threshold should not yet be treated as fixed.

7.3 A retraction

We had earlier suggested evaluating Tunnell’s coefficients for individual candidates via Hurwitz class numbers, avoiding the sieve’s dense-array memory requirement. This does not deliver the hoped-for improvement: evaluating one candidate this way requires $O(\sqrt{X/32})$ iterations, each with further work, making it roughly three orders of magnitude slower per candidate at the sieve’s target scale than the amortized $O(\log X)$ cost already achieved by the global FFT sieve. We withdraw this suggestion; the sieve’s memory-vs-time trade-off (Section 4) appears to need a genuinely different approach (e.g. a disk-backed or distributed segmentation), not a switch to per-candidate analytic evaluation.

[New in this revision] *Note added in this revision.* This retraction is directly relevant to one of the ideas considered for the present edition: a square-root-time, per-candidate Tunnell verification method of Bach and Ryan [10] was proposed as a possible memory-free alternative to our sieve. It is real, published mathematics, but it is in the same $O(\sqrt{n})$ -per-candidate complexity class as the Hurwitz class-number approach retracted above, and we would expect it to hit the same wall at our sieve’s target scale. It looks more promising as a fast, independent cross-check on the sieve’s (much smaller) set of survivors than as a wholesale replacement for the sieve itself.

7.4 A generator-free alternative: Strategy 2

[New in this revision]

Every method in §7.1–7.2 above works by pushing harder on `ellrank`’s generator search. Elkies–Goel’s own paper contains a second, structurally different method for exactly this situation – one that never calls `ellrank` at all, and so applies uniformly to the `partial`, `none`, and `skipped` (rank-4) categories of Table 3, not only to small hand-picked exceptional cases.

7.4.1 The Gaussian-integer parametrization

From $x^4 + y^4 = mz^2$ and $\gcd(x, y) = 1$, factor in $\mathbb{Z}[i]$:

$$x^2 + iy^2 = \beta w^2, \tag{16}$$

where $\beta \in \mathbb{Z}[i]$ has norm exactly m (one Gaussian prime above each rational prime dividing m , to the *first* power) and $w \in \mathbb{Z}[i]$ has norm z . For squarefree m with k distinct prime factors there are 2^{k-1} choices of β up to units and conjugation. For each choice this reduces the search for a solution to: find $w = p + ri$ of norm z (for some z) such that $\operatorname{Re}(\beta w^2)$ and $\operatorname{Im}(\beta w^2)$ are simultaneously perfect squares – a conic/Thue-type search using no elliptic-curve rank or generator information whatsoever.

Remark 7.1 (Corrected exponent). An earlier working note on this parametrization (and an early draft of the present revision process) used a *cubed* Gaussian prime, $\beta = \pi^3$. We have checked this against the exact reconstruction below and against Elkies–Goel’s own construction [3, §4]: the correct exponent for this general-purpose search is 1, not 3. The cubed-exponent identity is real but is a different, narrower lemma (the classical fact, going back to Lucas, that $x^4 + y^4 = m^3$ has no coprime solutions, already used in §1 above to fix the smallest admissible m) – not the general search tool.

7.4.2 Worked example, on the real N_1 solution

As a concrete check, we ran this pipeline on the known $m = 17$ solution (1). Since $17 = 4^2 + 1^2$, the two Gaussian primes above 17 are $4 + i$ and $4 - i$. Using exact integer arithmetic on $x = 427511122$, $y = 1322049209$:

- $x^2 + iy^2$ divides *exactly* by $4 + i$ (and not by its conjugate $4 - i$), confirming a well-defined single choice of β for $k = 1$;
- the quotient $q = (x^2 + iy^2)/(4 + i)$ has norm exactly z^2 , with $\sqrt{N(q)} = 426218494746902449 = z$ recovered exactly;
- q is itself an exact perfect square in $\mathbb{Z}[i]$: $q = (534805945 + 374434368i)^2$;
- reconstructing $\beta w^2 = (4 + i)(534805945 + 374434368i)^2$ reproduces $x^2 + iy^2$ exactly.

Every step of the pipeline is confirmed end to end on real data, not merely asserted.

Remark 7.2 (Scope of this validation). [\[New in this revision\]](#) To be precise about what has and has not been done: the check above validates the Strategy 2 pipeline on a single known case ($m = 17$). It has not yet been run, in either direction, on any of the currently-unresolved **partial/none/skipped** candidates – so while we are confident the method is correct and applicable, “applies to all such candidates” is a claim about the mathematics, not yet a report of a completed computation. Strategy 2 supplies a theoretically complete generator-free fallback based on the Gaussian-integer parametrization; its implementation has been validated on the known $m = 17$ solution but has not yet been deployed exhaustively across the unresolved candidate population.

7.4.3 A pipeline routed by outcome category

Strategy 2 needs no rank or generator information, so it directly targets exactly the three non-exhaustive outcome categories of Table 3, while Strategy 1 (`qfminim` on a saturated generator basis, now provably scoped via §5.4) is presumably still the cheaper option whenever a full basis is actually available:

Outcome category (Table 3)	Recommended method
full	Strategy 1: <code>qfminim</code> on the saturated basis
partial, none, skipped (rank 4)	Strategy 2: Gaussian-integer + conic search

Table 6: Proposed routing of candidates between the two search strategies.

Elkies–Goel implement the underlying rational-point search via a `qfsolve/hyperellratpoints`-style search on an auxiliary curve, with an induced height bound scaling like $N_{\max}^{1/8} m^{-1/8} C_i^{-1/4}$ for a curve-specific constant C_i , and search cost “essentially quadratic” in that height bound. This is a real cost – at our own $H = 160$ scale ($N_{\max} \sim e^{160}$) and mid-range m , the induced height bound is on the order of 10^7 – 10^8 (order-of-magnitude only; C_i needs computing per curve) – and should be benchmarked on the five known exceptional cases of §7.1 before being rolled out across the full `partial/none/skipped` population.

8 Parity and the Rank-Distribution Theorem

8.1 Every sieve-survivor has analytic rank at least 2 – an unconditional theorem

Every candidate m passing our sieve is, by Proposition 2.1, a product of primes $\equiv 1 \pmod{8}$; since such a product is again $\equiv 1 \pmod{8}$, this forces $m \equiv 1 \pmod{8}$ for every candidate, with no exceptions – verified directly across all 88,445 candidates in our sieve output. Consequently $n = 2m \equiv 2 \pmod{16}$ always, in particular $n \equiv 2 \pmod{8}$ always.

We initially established the consequence of this (Theorem 8.3 below) empirically, by testing the root number on 300 sampled candidates and finding it uniformly $+1$. That empirical finding can be replaced by an unconditional theorem, via the classical root-number formula of Birch and Stephens for exactly this family of curves.

Theorem 8.1 (Birch–Stephens, 1966 [1]). *Let n be a positive squarefree integer and $E_n : y^2 = x^3 - n^2x$ the associated congruent-number curve. The global root number is*

$$w(E_n/\mathbb{Q}) = \begin{cases} +1 & \text{if } n \equiv 1, 2, 3 \pmod{8}, \\ -1 & \text{if } n \equiv 5, 6, 7 \pmod{8}. \end{cases} \quad (17)$$

The root number is the product of local root numbers, $w(E_n/\mathbb{Q}) = \prod_{v \leq \infty} w(E_n/\mathbb{Q}_v)$. The archimedean factor is -1 for every $n > 0$. At an odd prime $p \nmid n$, E_n has good reduction and the local factor is $+1$; at odd $p \mid n$ (necessarily to the first power, since n is squarefree) the local factor is given by an explicit Jacobi-symbol formula depending only on $p \pmod{4}$, and the product of these over all $p \mid n$ collapses to a function of $n \pmod{8}$. The local factor at 2 is computed from the

Kodaira type of the minimal model at 2 (via the tables of Halberstadt or Rizzo in modern treatments); for squarefree n the possible types are limited and again resolve to a function of $n \bmod 16$. Combining the finitely many cases reproduces (17). The original 1966 argument of Birch and Stephens is phrased in terms of first 2-descents rather than local root numbers of the Weil–Deligne representation, but gives the same parity statement.

We independently tested (17) against PARI’s `ellrootno` across all six admissible residue classes mod 8 (four squarefree test values per class where possible, twenty-four curves total, including even n such as $n \equiv 2 \pmod{8}$, the case relevant here) before relying on it; every case matched the predicted sign exactly, with no exceptions.

Corollary 8.2. *For every $n \equiv 2 \pmod{8}$ – in particular for every $n = 2m$ arising from our sieve, since $m \equiv 1 \pmod{8}$ forces $n \equiv 2 \pmod{8}$ – $w(E_n/\mathbb{Q}) = +1$ unconditionally.*

Since the sign of the functional equation determines the parity of the order of vanishing of $L(E_n, s)$ at $s = 1$ ($w = +1 \iff$ even order), and Tunnell’s criterion holding is equivalent to $L(E_n, 1) = 0$ (order ≥ 1) by construction of the sieve, Corollary 8.2 upgrades our earlier empirical finding to an unconditional theorem:

Theorem 8.3. *Every candidate m surviving the sieve of Section 4 has $L(E_{2m}, s)$ vanishing to even order ≥ 1 at $s = 1$, hence to order ≥ 2 – unconditionally, via Theorem 8.1, with no appeal to computational sampling.*

This replaces the 300-candidate empirical check with a complete proof applicable to every candidate our sieve will ever produce, and gives an unconditional (not merely empirically-supported) consistency check on the whole pipeline: a reported algebraic rank of 0 or odd for a sieve survivor is necessarily an implementation error, not a legitimate outcome.

8.2 Correction: algebraic rank is not uniformly 2

An earlier version of this report, based on a smaller sample, claimed that every curve examined had algebraic rank exactly 2. This is incorrect, and we correct it here.

[New in this revision] *On the analytic/algebraic distinction.* Theorem 8.3 is a statement about *analytic* rank only: the order of vanishing of $L(E_{2m}, s)$ at $s = 1$. It does not, by itself, say anything about algebraic rank (the rank of $E_{2m}(\mathbb{Q})$ as an abelian group) – that link is the (unproved in general) Birch–Swinnerton-Dyer conjecture, which this paper does not assume. What *does* establish algebraic rank here is a separate, independent fact for each candidate: whenever `ellrank`’s 2-descent returns *matching* lower and upper bounds (e.g. $[2, 2]$ or $[4, 4]$ in Table 1 and throughout), the algebraic rank is pinned down unconditionally by that descent computation alone, with no BSD or analytic input required – the lower bound comes from exhibited independent points, the upper bound from the Selmer computation, and equality forces the true rank to sit exactly between them. Theorem 8.3 is used only as an unconditional *consistency check* on this independent computation (a reported rank of 0 or odd would flag an implementation bug), not as a step in establishing the algebraic rank itself. With this distinction stated explicitly: Theorem 8.3 establishes that *analytic* rank is at least 2 and even; direct 2-descent computations (`ellrank`, independently, per candidate, with matching bounds) establish that *algebraic* rank is at least 2 and even for every candidate tested, and exactly 4 for a specific measured subset – it does not exclude algebraic rank 6, 8,

etc. Direct diagnosis (Section 5.3) found 3,611 of 88,445 candidates (4.08%) to have algebraic rank exactly 4, confirmed via `ellrank` returning matching bounds $[4, 4]$ in every case, with no exceptions and no ambiguity in the descent.

Proposition 8.4. *Among the 88,445 sieve-surviving candidates $m \leq 8.8 \times 10^7$: 4.08% have algebraic rank 4; the remainder have rank 2 (no rank-0 or rank-odd candidate was found anywhere in this investigation, as guaranteed unconditionally by Theorem 8.3; no rank ≥ 6 candidate was found either, though higher ranks were not specifically searched for and are not excluded by anything proven here).*

This has a practical consequence for pipeline design: Theorem 8.3 justifies skipping a rank lower-bound check (rank ≥ 2 is guaranteed a priori, unconditionally) and gives a proven, not merely empirical, sanity check – a reported rank of 0 or odd rank for a sieve-surviving candidate would indicate an implementation bug. It does not, however, justify assuming rank exactly 2 and building a fixed-dimension search around that assumption, as our own earlier draft implicitly did by only implementing the rank-1 and rank-2 cases of `qfminim` search and silently skipping the 4.08% of candidates with rank 4 – a gap that §7.4’s generator-free method now closes.

9 Discussion and Limitations

- The rigorous, honestly-scoped coverage of Proposition 6.1 is a small fraction of the full 88,445-candidate pool, which is itself a small fraction ($< 0.1\%$) of the $\sim 2.2 \times 10^{11}$ figure quoted in §4.4 as the scale a from-scratch resolution would require – a figure we now flag (§4.4, footnote) as itself needing an explicit derivation. Per §1, an unconditional resolution of the minimality question already exists independently, via [3], regardless of this figure.
- The sieve’s memory requirement remains the binding constraint on extending our own candidate pool further (Section 4); the 9–10 TB estimate for the full range was not tested directly, and §4.4 now points to an existing computation at a larger scale that may make a from-scratch rewrite unnecessary.
- `ellrank`’s reliability for m well beyond 8.8×10^7 has not been assessed.
- The height bound $H = 160$ was calibrated empirically (three points) in the original edition; §5.4 supplies an unconditional, model-specific alternative, and §5.5 (**[New in this revision]** new in this revision) connects it rigorously to N , which lowers the honestly-provable N -bound from the originally-claimed 1.5×10^{69} to 1.56×10^{58} uniformly (more for individual small m). The two-sided inequality of §5.4 has only been checked numerically (20,000 random cases) and symbolically for its hardest step, not machine-verified end to end.
- **[New in this revision]** The full classification (Table 3) checks generator *count* against proven rank, but – as made explicit in the new remark closing §5.3 – does not check generator *saturation*. Every count in this paper attached to the word “exhaustive” or “full” should be read subject to this caveat until `ellsaturation` certificates are produced; see §9.1(a) below, now the top-priority item.
- **[New in this revision]** Relatedly, we have not confirmed whether the `qfminim` search enumerates all four torsion cosets of the free lattice or only the identity coset (§5.2, Remark 5.1); this is a second, independent question from saturation and is likewise unresolved.

- The triage rule of Section 7.2 rests on six data points and an approximation (dropped cross term, $\#Sha = 1$ assumed); a specific numerical threshold should be treated as provisional.
- The rank-4 finding (Section 8.2) itself came from re-diagnosing a bug rather than from a planned investigation – a reminder that claims of computational uniformity warrant deliberate adversarial checking, not just repeated confirmation on convenient samples.
- The survey of Section 10 used a single fixed height bound $H = 10^4$ throughout; some of the 16,397 candidates searched there may have additional solutions beyond that bound (as $m = 17$ itself did beyond the earlier $H = 160$), and the 96% of candidates in $[10^6, 10^7]$ without a full generator basis remain entirely unsearched by this method – though now, in principle, reachable via §7.4’s Strategy 2. **[New in this revision]** Its two reported ratios should also be read per the caveat added to §10.1: the solution-density figure is conditional on the (non-random) fully-certified subset, not a population-level estimate.
- **[New in this revision]** The FFT sieve of §4 is validated by spot-checks against brute force, not by a global numerical-error proof; an exact multimodular/NTT sieve (in the spirit of [6], §4.4) or an independent modular cross-check would upgrade this to a certified computation.

9.1 Future work

- (a) **[New in this revision]** *Saturation certification* (top priority). Run `ellsaturation` on every full-classified candidate (§5.3) and record the resulting index bound; without this, “full” means exhaustive over the sublattice found, not over $E_m(\mathbb{Q})$ itself. A complete treatment would also report the 2-Selmer dimension and torsion subgroup alongside rank and generators for each candidate, especially the rank-4 population, as a full descent certificate rather than a rank bound alone.
- (b) **[New in this revision]** *Torsion-coset confirmation*. *Status: new*. Confirm and state explicitly whether all four torsion cosets are searched by the current `qfminim` pipeline (§5.2, Remark 5.1), or give an argument for why only the identity coset can contain an admissible primitive solution.
- (c) A segmented, disk-backed, or distributed sieve implementation, given per-candidate analytic evaluation is confirmed not to help (Section 7). *Status: redirected*. Rather than building this from scratch, §4.4 identifies an existing computation (Hart–Tornaríá–Watkins [6]) at a larger scale using essentially this design; obtaining or benchmarking against it is likely more efficient than an independent rewrite. A fully certified version would use an exact multimodular/NTT final sieve rather than floating-point FFT.
- (d) Completing the fixed-code rigorous re-verification across the remaining 86,984 candidates, now understood to require saturation certificates (item (a)) to count as genuinely exhaustive.
- (e) *A genuinely different technique for the generator-finding-hard cases and for the 4.08% rank-4 population*. *Status: addressed in principle, not yet at scale*. §7.4 supplies this via Elkies–Goel’s own generator-free Strategy 2, verified end to end on the real N_1 solution (Remark 7.2 states this scope explicitly); benchmarking and deploying it at scale on the `partial/none/skipped` population remains open. **[New in this revision]** A further-reaching goal, not yet attempted, would be a genuine completeness *theorem* for Strategy 2 – proving completeness of

the set of Gaussian-integer factors and conic parametrizations used, with explicit parameter-height bounds and exact recovery of (x, y, z) – rather than the case-by-case validation given here.

- (f) *A rigorous, model-specific height bound in place of the empirical calibration. Status: addressed, and connected to an N -bound.* §5.4 supplies the height bound and §5.5 ([New in this revision] new) supplies the missing bridge to N , correcting the resulting figure. [New in this revision] A further sharpening, not yet attempted here, would derive the height comparison through the explicit 2-isogeny to Elkies–Goel’s own model [3, §3] rather than redoing the gcd-control argument natively on E_m ; this canonical-height work may be one of this paper’s more durable contributions and would repay being developed further in its own right.
- (g) Validating the triage rule against a larger sample.
- (h) Determining whether rank exceeds 4 for any candidate in this family, and whether the 4.08% rank-4 proportion is itself asymptotically stable as m grows.
- (i) *A theoretical explanation for the sharp decay of both the generator-availability rate and the solution-density rate as $m \rightarrow 10^7$, and for why $m = 17$ is such an outlier. Status: partially addressed.* The paper’s own regulator-based triage rule (§7.2) already supplies a plausible mechanism, and we make it explicit here rather than leaving it implicit: if $\text{Reg}(E_m)$ grows on average with m (as Table 5’s own three-decade spread of estimated second-generator heights already suggests), then by a Minkowski-type bound for a rank-2 lattice the height of the *shortest* non-torsion vector should grow roughly like $\sqrt{\text{Reg}(E_m)}$, directly explaining why both a fixed `ellrank` effort (§10.1’s collapsing full-basis rate) and a fixed search height $H = 10^4$ (§10.1’s collapsing solution density) succeed less and less often as m grows. On this account $m = 17$ is an outlier simply because, being the smallest admissible m , it has by far the smallest typical regulator in the family – exactly as observed directly in Table 5 ($h(G_1) = 2.51$ for $m = 17$ against 4.4–8.8 for the next five smallest admissible values tested). This is a heuristic account consistent with all data gathered so far, not a proof, and we list it as only *partially* addressed for that reason.
- (j) [New in this revision] *Numerical certification of the lattice enumeration.* `qfminim` operates on a floating-point Gram matrix; a fully certified exhaustiveness claim would document the working precision and either verify stability under increased precision or use interval arithmetic / rigorous enclosures for the height-pairing matrix, rather than trusting default precision throughout.
- (k) [New in this revision] *Reproducibility archive.* Publish a versioned, public repository containing the sieve and search source code, exact environment and PARI/GP/compiler versions and command lines, the full candidate list, rank/saturation/`qfminim` outputs, the complete solution catalogue, and checksums – beyond the JSON result logs already described in §11.

10 A Height-Bounded Survey of Solution Density for $m \leq 10^7$

Motivated by the discovery that $m = 17$ admits more solutions than the two originally known once the search height is increased (Section 5.2 used $H = 160$; N_1, N_2 are only the two smallest), we

conducted a systematic survey: for every sieve-surviving $m \in [17, 10^7]$ with a full 2-generator basis (Section 5.3), we ran the exhaustive `qfminim` search at a fixed height bound $H = 10^4$ – roughly 60–27 times the height of N_1, N_2 respectively, and comfortably above every previously-known solution in this family. Candidates without a full basis were left unsearched, as in Section 6.

10.1 Coverage and results by range

Range of m	Candidates	Full-basis rate	Solutions found	Curves with ≥ 1 solution
$[17, 10^4]$	72	74% (53/72)	111	12/53
$[10^4, 10^5]$	385	32% (122/385)	2	2/122
$[10^5, 10^6]$	2,259	12% (267/2259)	0	0/267
$[10^6, 10^7]$	13,681	4.0% (546/13681)	0	0/546
Total	16,397	—	113	14

Table 7: Coverage and results of the $H = 10^4$ survey, by range of m . “Full-basis rate” is the fraction of candidates for which `ellrank` returned a complete 2-generator basis, making the `qfminim` search genuinely exhaustive for that candidate; the remainder were left unsearched.

Two monotonic trends are apparent across five orders of magnitude of m :

- The full-basis rate collapses: $74\% \rightarrow 32\% \rightarrow 12\% \rightarrow 4.0\%$, roughly halving or worse with each decade of m . This is consistent with, and adds further evidence for, the regulator-based explanation of Section 7.2 (see also §9.1(g)): missing generators typically have height growing with m , so `ellrank`’s fixed-effort search finds a complete basis ever less often.
- Solution density collapses even faster: among the (shrinking) set of fully-searchable candidates, the fraction with any solution at all drops from 12/53 (23%) to 2/122 (1.6%) to 0/267 to 0/546. No curve in $[10^5, 10^7]$ produced a single solution up to $H = 10^4$, despite 813 curves being exhaustively searched there.

Remark 10.1 (These are conditional, not population, frequencies). [\[New in this revision\]](#) The two ratios above should not be read as independent measurements of an underlying solution density among *all* sieve survivors in each range. The full-basis rate figure already reports the correct quantity for its own question (fully-certified curves out of all sieve survivors in that range). The second figure, however – the solution-density fraction – is computed only over the fully-certified *subset*, which Section 7.2’s regulator-based explanation gives us reason to think is a systematically biased sample (curves with smaller-than-typical regulators, hence smaller-than-typical generator heights, are exactly the curves `ellrank` is more likely to fully resolve). We therefore report the two ratios separately and explicitly, as

$$\frac{\#\{\text{fully certified curves}\}}{\#\{\text{sieve survivors}\}} \quad \text{and} \quad \frac{\#\{\text{solutions found}\}}{\#\{\text{fully certified curves}\}},$$

rather than combining them into a single “solution density” figure for the range, and caution against extrapolating the second ratio to the (systematically different) population of candidates that `ellrank` could not fully resolve.

Remark 10.2. Fourteen distinct values of m produced at least one solution: 17, 97, 113, 193, 257, 337, 433, 577, 641, 673, 2833, 4481, 20513, 62033. Of these, $m = 17$ is a marked outlier, contributing

86 of the 113 solutions found (76%); the next richest, $m = 113$, contributes 7. Five values of m (337, 433, 577, 641, 673) have exactly one solution each, all substantially larger than N_1 .

10.2 Complete solution data

All 113 solutions were independently verified (fresh computation of $x^4 + y^4$, $\gcd(x, y) = 1$, $m \mid z$, and $x^4 + y^4 = m^3(z/m)^2$, i.e. $N = a^2b^3$ with $b = m$, $a = z/m$) before inclusion. The ten smallest, by canonical height, are given explicitly below; the complete set of 113 (with canonical height and decimal size only, to keep this document a reasonable length) is in Table 8 at the end of this document, and full (x, y, z, a, b) values for all 113 are in the accompanying data file `full_solution_data_with_heights.json`.

$m = 17$, $\hat{h} = 84.11$, **37 digits** ($= N_1$):

$x = 427511122$, $y = 1322049209$

$m = 17$, $\hat{h} = 138.44$, **61 digits** ($= N_2$):

$x = 572132418369898$, $y = 988478679472373$

$m = 17$, $\hat{h} = 240.37$, **105 digits**:

$x = 72786971915852912664451837$, $y = 112681733207155381861368622$

$m = 113$, $\hat{h} = 372.89$, **162 digits**:

$x = 13328234848468028322634068228462251903168$,

$y = 29121398138510909704161998188666074519659$

$m = 17$, $\hat{h} = 403.38$, **176 digits**:

$x=23330982316596035764140805788284142220333873$, $y=60328420548951109551618971740883735652479494$

$m = 97$, $\hat{h} = 547.04$, **238 digits**:

$x=128354855299088400081857716533235557180839113089775198820998$, $y=228976722264547978356114923500890684931693170291166378243569$

$m = 17$, $\hat{h} = 607.23$, **264 digits**:

$x=21755462090284929768037355474417812356239609310901316280550332287$, $y=835849626760004699936539489265565354008520177638124232352976068906$

$m = 17$, $\hat{h} = 709.39$, **309 digits**:

$x=7697390131777063299464087904649067878470635143928099060152150706425544248154$, $y=104803531922643718187890183824853555677518811837843082631389431005526055193101$

$m = 17$, $\hat{h} = 756.98$, **329 digits**:

$x=11523685497451073207491354768893723631841448494274575963336288968979444006793987353$, $y=12230041453090614020527347549049806539638606678166471619982320912779108191635690806$

$m = 17$, $\hat{h} = 872.40$, **379 digits**:

$x=37181074309366199544314162510584309484564461908613803393487720006570875829775501365533693991202$, $y=43172754398208284831194521827216175611511355406269413907294425781081462686754869074543389031041$

For each of the above, $b = m$ and $a = z/m$ recover the $N = a^2b^3$ powerful decomposition directly; e.g. for the smallest (N_1), $a = 25071676161582497$, $b = 17$. All ten were independently re-verified (fresh $x^4 + y^4$ computation, $\gcd(x, y) = 1$, $17 \mid z$ or $113 \mid z$ as appropriate, and the $N = a^2b^3$ structure) before inclusion here.

11 Reproducibility

All code, intermediate data, and checkpoint files are available alongside this document: the FFT sieve (both implementations discussed in Section 4); the PARI/GP interface for rank, generator, and lattice-point computation, including the corrected generator-count check of Section 5.3; the descent map with its validation tests (including the corrected form of §3); the complete $H = 10^4$ survey of Section 10 (`full_solution_data_with_heights.json`, plus per-range checkpoint logs); [\[New in this revision\]](#) the symbolic (computer-algebra) derivation and the 20,000-case numerical stress test underlying the height bound of §5.4; [\[New in this revision\]](#) the exact-arithmetic Gaussian-integer reconstruction of §7.4 on the real N_1 solution; and complete result logs (JSON Lines) sufficient to independently verify every count in this report, including the pre-fix and post-fix result sets side by side.

A Complete Solution Data

A.1 $m = 17$

$$\begin{aligned} N_1 &= 427511122^4 + 1322049209^4 = 3088257489493360278725196965477359217 \\ &= 17^3 \cdot 73993169^2 \cdot 338837713^2 \\ N_2 &= 572132418369898^4 + 988478679472373^4 \\ &= 1061853595348370798528584585707993395597624934311961270177857 \end{aligned}$$

A.2 $m = 113$

$$\begin{aligned} x &= 13328234848468028322634068228462251903168 \\ y &= 29121398138510909704161998188666074519659 \\ z &= 81509829735780805715967239211386772518093334385069631637723303670954268130961057 \\ x^4 + y^4 &= 7507553148218253842934944945095100830397379550839941966441991626167799290502535671876 \\ &26522281487324180008794315593453957534795502257893302270660768600912969137 \quad (162 \text{ digits, verified} \\ &\text{by direct computation.}) \end{aligned}$$

A.3 $m = 2833$

x and y below are each broken across lines purely for typesetting.

$$\begin{aligned} x &= 139264853254157834767954391058677749547159139409736771861326052870713885593456724289235449 \\ &25803667278045381136625428060483629204081185241364615429172 \end{aligned}$$

$$\begin{aligned} y &= 255575393430652376938200669229598601481448813777448108661734531814577934605554706611148638 \\ &01120689323631814634942252269774624097905010233413576576263 \end{aligned}$$

z and $x^4 + y^4$ (594 digits) are provided in the accompanying data file (record for $m = 2833$).

B Complete $H = 10^4$ Survey Results

Table 8 lists all 113 solutions found in the survey of Section 10, sorted by canonical height, giving m , canonical height $\hat{h}$, and the decimal digit count of $N = x^4 + y^4$ for each. Full (x, y, z, a, b) values for every row are in the accompanying file `full_solution_data_with_heights.json`.

Table 8: All 113 solutions found in the $H = 10^4$ survey (Section 10), sorted by canonical height.

m	Canonical height $\hat{h}$	Decimal digits of N
17	84.11	37
17	138.44	61
17	240.37	105
113	372.89	162
17	403.38	176
97	547.04	238
17	607.23	264
17	709.39	309
17	756.98	329
17	872.40	379
17	878.91	382
17	1015.17	441
17	1184.68	515
17	1246.00	542
2,833	1367.18	594
17	1483.95	645
17	1565.03	680
17	1830.20	795
17	1953.20	849
17	1972.74	857
17	2014.28	875
17	2102.72	914
17	2163.33	940
17	2285.96	993
17	2461.76	1070
17	2462.83	1070
193	2518.62	1094
17	2625.00	1141
17	2768.24	1203
337	2806.91	1219
17	2971.39	1291
257	3019.81	1312
17	3033.54	1318
17	3053.31	1326
97	3064.22	1331
113	3356.04	1458
97	3368.38	1463
17	3461.11	1503
433	3496.92	1519
20,513	3520.41	1529
17	3569.08	1550
17	3630.39	1577

m	Canonical height $\hat{h}$	Decimal digits of N
17	3814.85	1657
17	3829.04	1663
17	3869.89	1681
17	3998.80	1737
4,481	4078.42	1772
17	4121.33	1790
17	4153.89	1805
17	4180.63	1816
17	4364.58	1896
17	4379.14	1902
17	4583.37	1991
17	4806.75	2088
17	4846.38	2105
113	4877.61	2119
17	4887.00	2123
97	4923.34	2139
17	4970.09	2159
193	5002.46	2173
17	5256.01	2283
17	5465.06	2374
113	5469.02	2376
17	5478.64	2380
17	5600.48	2433
17	5771.63	2507
62,033	5965.32	2591
17	6009.25	2610
641	6010.51	2611
17	6139.24	2667
17	6272.97	2725
17	6316.67	2744
17	6384.50	2773
17	6415.51	2787
17	6459.44	2806
17	6662.92	2894
257	6739.33	2927
17	6783.77	2947
17	6803.31	2955
17	6812.81	2959
17	6986.83	3035
97	7136.35	3100
17	7151.94	3107
17	7230.92	3141
17	7233.07	3142
113	7269.35	3157
17	7376.78	3204
577	7376.81	3204
17	7422.35	3224
17	7851.56	3410
673	7853.37	3411
17	7910.16	3436
17	8006.51	3478
97	8048.85	3496

m	Canonical height $\hat{h}$	Decimal digits of N
17	8151.34	3540
17	8154.62	3542
17	8338.57	3622
17	8537.49	3708
17	8719.19	3787
17	8761.77	3806
193	8937.16	3882
113	9043.58	3928
17	9046.05	3929
17	9071.95	3940
17	9136.50	3968
113	9322.33	4049
17	9439.85	4100
17	9443.07	4101
17	9477.17	4116
17	9619.57	4178
17	9721.87	4223
17	9757.94	4238
17	9882.59	4292

C Derivation Details for the Height Bound of Section 5.4

This appendix records the intermediate steps behind Section 5.4 in full, for independent checking.

The duplication polynomials

For $E_m : v^2 = u^3 + Au$ with $A = -4m^2$, the standard duplication formula for the x -coordinate, $x([2]P) = \frac{x^4 - 2Ax^2 - 8Bx + A^2}{4(x^3 + Ax + B)}$ (with $B = 0$), specializes to

$$\varphi(x) = x^4 + 8m^2x^2 + 16m^4 = (x^2 + 4m^2)^2, \quad \psi(x) = 4x^3 - 16m^2x.$$

Both were checked by direct symbolic expansion against the general duplication formula (computer algebra), not assumed.

The Bézout identity

Since φ has no real roots and ψ 's roots $(0, \pm 2m)$ are real, φ and ψ are coprime in $\mathbb{Q}(m)[x]$. Running the extended Euclidean algorithm (verified symbolically) gives

$$F(x)\varphi(x) + G(x)\psi(x) = 1, \quad F(x) = \frac{16m^2 - 3x^2}{256m^6}, \quad G(x) = \frac{x(20m^2 + 3x^2)}{1024m^6}.$$

Clearing denominators, $R \cdot F, R \cdot G \in \mathbb{Z}[m][x]$ for $R = 1024m^6$: explicitly,

$$1024m^6 \cdot F(x) = 64m^2 - 12x^2 \in \mathbb{Z}[m][x], \quad 1024m^6 \cdot G(x) = 20m^2x + 3x^3 \in \mathbb{Z}[m][x].$$

The gcd bound

Because φ is monic, the standard lemma (used identically by Elkies–Goel [3, Lemma 1]) states: for $u = p/q$ in lowest terms, writing $\Phi(p, q) = q^4\varphi(p/q) = (p^2 + 4m^2q^2)^2$ and $\Psi(p, q) = q^4\psi(p/q) = 4pq(p^2 - 4m^2q^2)$ for the homogenized numerator and denominator of the doubled coordinate,

$$\gcd(\Phi(p, q), \Psi(p, q)) \mid R = 1024m^6.$$

Two-sided height estimate

Upper bound. Using $p^2, q^2 \leq H(u)^2$ (with $H(u) = \max(|p|, |q|)$): $|\Phi| = (p^2 + 4m^2q^2)^2 \leq (1 + 4m^2)^2 H(u)^4$ and $|\Psi| \leq 4(1 + 4m^2)H(u)^4$; for $m \geq 1$ the first dominates, giving $\max(|\Phi|, |\Psi|) \leq (1 + 4m^2)^2 H(u)^4$, hence (since reduction can only shrink the height)

$$h([2]P) \leq 4h(u) + 2\ln(1 + 4m^2).$$

Lower bound. Since $p^2 + 4m^2q^2 \geq \max(p^2, 4m^2q^2)$ and, case-checking whichever of $|p|, |q|$ attains $H(u)$, one has $p^2 + 4m^2q^2 \geq H(u)^2$ for all $m \geq 1$; hence $|\Phi| \geq H(u)^4$, so $\max(|\Phi|, |\Psi|) \geq H(u)^4$. Combined with the gcd bound above,

$$h([2]P) \geq 4h(u) - \ln(1024m^6) = 4h(u) - 6\ln m - 10\ln 2.$$

Together these give equation (11) of the main text. Telescoping with $\sum_{k \geq 1} 4^{-k} = 1/3$ gives equation (12).

Numerical stress test

The gcd-divisibility claim and the two-sided per-doubling inequality (11) were tested across 20,000 independently-sampled triples (m, p, q) with $m \in \{17, 41, 73, 89, 97, 113, 4721, 100003\}$ and p, q random up to 10^6 with $\gcd(p, q) = 1$: zero violations of either the gcd claim or either side of (11), with the tightest observed margins 12.7 (lower bound) and 1.6×10^{-13} (upper bound) – the latter indicating the upper-bound derivation above is close to sharp, which is evidence against an algebraic error (a genuine error would more plausibly produce either an outright violation or unstructured slack, not a bound realized to 13 decimal digits).

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